

Dokumentowe bazy danych – MongoDB

Ćwiczenie 1

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Test połączenia

- informacja o wersji

```
db.version();
```

- wynik

```
{
  "result": "7.0.7"
}
```

- lista baz danych na serwerze

```
show dbs;
show databases;
db.adminCommand('listDatabases');
```

- wynik

```
[
  {
    empty: false,
    name: "admin",
    sizeOnDisk: 40960,
  },
  {
    empty: false,
    name: "config",
    sizeOnDisk: 98304,
  },
  {
    empty: false,
    name: "local",
    sizeOnDisk: 40960,
  },
];
```

- wybór bazy danych
 - baza o nazwie univ

```
use univ;
```

- informacja o bieżącej bazie danych

```
db;
```

- informacja o kolekcjach

```
show collections;
db.getCollectionNames();
```

Proste operacje/zapytania

- MongoDB simple query
 - <https://www.mongodb.com/docs/manual/tutorial/query-documents/>
-

Przykład 1

proste operacje na dokumentach

- wstaw/stwórz pierwszy dokument
 - w tym momencie tworzona jest baza danych i kolekcja (jeśli wcześniej nie istniały)

```
db.student.insertOne({
  student_id: 1,
  firstname: "John",
  lastname: "Gold",
  age: 25,
});
```

- sprawdź listę baz i kolekcji

```
show databases;
```

- wynik

```
[
  {
    empty: false,
    name: "admin",
    sizeOnDisk: 40960,
  },
  {
    empty: false,
    name: "config",
    sizeOnDisk: 98304,
  },
  {
    empty: false,
    name: "local",
    sizeOnDisk: 40960,
  },
  {
    empty: false,
    name: "univ",
    sizeOnDisk: 8192,
  },
];
```

```
show collections;
```

- wynik

```
[
  {
    badge: "",
    name: "student",
  },
];
```

- wyszukaj dokumenty w kolekcji `employees`

```
db.student.find();
db.student.find({});
```

- wynik

```
[
  {
    _id: { $oid: "67e86e45b1c69e3d7e42c9c0" },
    age: 25,
    firstname: "John",
    lastname: "Gold",
    student_id: 1,
  },
];
```

- stwórz indeks zapewniający unikalność atrybutu `"student_id"`

```
// tworzenie indeksu
db.student.createIndex({ student_id: 1 }, { unique: true });

// inf. o indeksach w kolekcji student
db.student.getIndexes();

// usunięcie indeksu o nazwie
db.student.dropIndexes("student_id_1");
```

- wstaw kilka dokumentów do kolekcji `student`
 - np

```
db.student.insertOne({
  student_id: 2,
  firstname: "James",
  lastname: "Bond",
});

db.student.insertOne({
  firstname: "John",
  lastname: "Bond",
});
```

- wyszukaj dokumenty w kolekcji `student`

```
db.student.find();
db.student.find({});
db.student.find({ firstname: "John" });
db.student.find({ student_id: 1 });

db.student.find({ _id: ObjectId("67e669647013d10c86e71c87") });
```

- zmodyfikuj wybrane dokumenty
 - po wykonaniu każdego z przykładów sprawdź wynik za pomocą `db.student.find()`
- np

```
db.student.updateOne(
  { student_id: 1 },
  {
    $set: {
      firstname: "Adam",
      lastname: "Silver",
    },
  },
);
```

```
db.student.updateOne(
  { student_id: 1 },
  {
    $inc: { age: 5 },
  },
);
```

```
db.student.updateOne(
  { student_id: 1 },
  {
    $set: {
      firstname: "John",
      lastname: "Gold",
    },
    $inc: {
      age: -2,
    },
  },
);
```

```
db.student.updateOne(
  { student_id: 2 },
  {
    $set: {
      age: 19,
    },
  },
);
```

```
},
);
```

```
db.student.updateOne(
  { student_id: 2 },
  {
    $unset: {
      age: "",
    },
  },
);
```

- usunąć wybrane dokumenty
 - po wykonaniu każdego z przykładów sprawdź wynik za pomocą `db.student.find()`
- np

```
db.student.deleteOne({ student_id: 1 });
db.student.deleteMany({ student_id: 1 });
db.student.deleteOne({ _id: ObjectId("67e669647013d10c86e71c87") });
```

- `replace`

```
db.student.find({"student_id": 1});
```

```
db.student.replaceOne(
  {
    "student_id": 1
  },
  {
    "student_id": 1,
    "firstname": "Jan",
    "lastname": "Kowalski",
    "hobby": "Ski"
  }
);
```

- wynik

```
[
  {
    "_id": {"$oid": "67ec1a5b5d92f33cd84a2598"},
    "hobby": "Ski",
    "lastname": "Kowalski",
    "firstname": "Jan",
    "student_id": 1
  }
]
```

- tablice
 - `grades` - tablica ocen studenta

```
db.student.updateOne(
  {
    student_id: 1,
  },
  {
    $set: {
      grades: [3, 4, 5, 6, 8],
    },
  },
);
```

```
db.student.find({ student_id: 1 });
```

- wynik

```
[
  {
    "_id": {"$oid": "67ec1a5b5d92f33cd84a2598"},
    "firstname": "Jan",
```

```

    grades: [3, 4, 5, 6, 8],
    hobby: "Ski",
    lastname: "Kowalski",
    student_id: 1,
  },
];

```

- średnia ocena

```

db.student.aggregate([
  {
    $match : {
      "student_id": 1
    }
  },
  {
    $project: {
      "firstname": 1,
      "lastname": 1,
      "student_id": 1,
      "grades": 1,
      "averagegrade": { $avg: "$grades" }
    }
  }
])

```

- wynik

```

[
  {
    "_id": {"$oid": "67ec1a5b5d92f33cd84a2598"},
    "averagegrade": 5.2,
    "firstname": "Jan",
    "grades": [3, 4, 5, 6, 8],
    "lastname": "Kowalski",
    "student_id": 1
  }
]

```

- dodanie elementu do tablicy

```

db.student.updateOne(
  { student_id: 1 },
  { $push: { grades: 10 } }
);

```

```

db.student.find({"student_id": 1});

```

- wynik

```

[
  {
    _id: { $oid: "67ec1a5b5d92f33cd84a2598" },
    firstname: "Jan",
    grades: [3, 4, 5, 6, 8, 10],
    hobby: "Ski",
    lastname: "Kowalski",
    student_id: 1,
  },
];

```

- modyfikacja elementu w tablicy
 - indeksy zaczynają się od 0

```

db.student.updateOne({ student_id: 1 }, { $set: { "grades.2": 12 } });

```

```

db.student.find({"student_id": 1});

```

- wynik

```
[
  {
    _id: { $oid: "67ec1a5b5d92f33cd84a2598" },
    firstname: "Jan",
    grades: [3, 4, 12, 6, 8, 10],
    hobby: "Ski",
    lastname: "Kowalski",
    student_id: 1,
  },
];
```

- `$map` - wykonanie operacji dla każdego elementu tablicy
 - <https://www.mongodb.com/docs/manual/reference/operator/aggregation/map/>
 - w przykładzie poniżej
 - dzielimy każdą ocenę przez 20

```
db.student.updateOne({ student_id: 1 }, [
  {
    $set: {
      grades: {
        $map: {
          input: "$grades",
          as: "g",
          in: { $divide: ["$$g", 20] },
        },
      },
    },
  },
]);
```

```
db.student.find({ student_id: 1 });
```

- wynik

```
[
  {
    _id: { $oid: "67ec1a5b5d92f33cd84a2598" },
    firstname: "Jan",
    grades: [0.15, 0.2, 0.6, 0.3, 0.4, 0.5],
    hobby: "Ski",
    lastname: "Kowalski",
    student_id: 1,
  },
];
```

Zadanie 1

- Wykonaj kilka własnych eksperymentów z operacjami CRUD
- przydatne linki
 - <https://www.mongodb.com/docs/manual/crud/>
 - <https://www.mongodb.com/docs/manual/reference/method/db.collection.find/>
 - <https://www.mongodb.com/docs/manual/reference/method/db.collection.insertOne/>
 - <https://www.mongodb.com/docs/manual/reference/method/db.collection.updateOne/>
 - <https://www.mongodb.com/docs/manual/reference/method/db.collection.deleteOne/>

Zadanie 1 - rozwiązanie

Wyniki:

przykłady, kod, zrzuty ekranów, komentarz ...

```
use bank;

// CREATE
db.accounts.insertOne(
  {
    _id: 0,
    name: "PL1234567890",
    balance: NumberDecimal(999),
    owner_id: 1
  }
)

db.accounts.insertOne(
```

```
{
  _id: 2,
  name: "PL8273645210",
  balance: 2,
  owner_id: 1
}
)

db.accounts.createIndex({name:1}, {unique: true})

db.customers.insertOne(
{
  _id: 1,
  firstname: "Jan",
  lastname: "Kowalski",
  accounts_id: [1,2]
}
)

// READ
db.accounts.find(
{
  _id:0,
  balance:999
}
)

db.customers.find(
{
  firstname: "Jan"
}
)

db.accounts.find(
{
  name: "PL8273645210"
}
)

// UPDATE
db.accounts.updateOne( {_id: 0}, {"$set": {balance: NumberDecimal(24)}} )
db.customers.replaceOne( {_id: 1},
{
  _id: 1,
  firstname: "Marek",
  lastname: "Nowak",
  accounts_id: []
})

// DELETE
db.accounts.deleteOne({_id:0})
db.customers.deleteMany({})
```

Przykład 2

- wybierz bazę `employees`

```
use employees;

db;
```

- wstaw dokument do kolekcji `employees`

```
db.employees.insertOne({
  EmployeeID: 1,
  FirstName: "Nancy",
  LastName: "Davolio",
  Address: {
    Street: "507 - 20th Ave. E. Apt. 2A",
    City: "Seattle",
    Country: "USA",
  },
  Title: "Sales Representative",
  TitleOfCourtesy: "Ms.",
  BirthDate: ISODate("1948-12-08T00:00:00.000Z"),
  HireDate: ISODate("1992-05-01T00:00:00.000Z"),
  Phone: ["(206) 555-9857", "(206) 555-9858"],
  Salary: 1000,
});
```

- sprawdź zawartość kolekcji

```
db.employees.find();
```

- wstaw kilka kolejnych dokumentów za pomocą `insertMany`

```
db.employees.insertMany([
  {
    EmployeeID: 2,
    FirstName: "Andrew",
    LastName: "Fuller",
    Address: {
      Street: "908 W. Capital Way",
      City: "Tacoma",
      Country: "USA",
    },
    Title: "Vice President, Sales",
    TitleOfCourtesy: "Dr.",
    BirthDate: ISODate("1952-02-19T00:00:00.000Z"),
    HireDate: ISODate("1992-08-14T00:00:00.000Z"),
    Phone: ["(206) 555-9482"],
    Salary: 10000,
  },
  {
    EmployeeID: 3,
    FirstName: "Janet",
    LastName: "Leverling",
    Address: {
      Street: "722 Moss Bay Blvd.",
      City: "Kirkland",
      Country: "USA",
    },
    Title: "Sales Representative",
    TitleOfCourtesy: "Ms.",
    BirthDate: ISODate("1963-08-30T00:00:00.000Z"),
    HireDate: ISODate("1992-04-01T00:00:00.000Z"),
    Phone: ["(206) 555-3412"],
    Salary: 1200,
  },
  {
    EmployeeID: 4,
    FirstName: "Margaret",
    LastName: "Peacock",
    Address: {
      Street: "4110 Old Redmond Rd.",
      City: "Redmond",
      Country: "USA",
    },
    Title: "Sales Representative",
    TitleOfCourtesy: "Mrs.",
    BirthDate: ISODate("1937-09-19T00:00:00.000Z"),
    HireDate: ISODate("1993-05-03T00:00:00.000Z"),
    Phone: ["(206) 555-8122"],
    Salary: 1100,
  },
  {
    EmployeeID: 5,
    FirstName: "Steven",
    LastName: "Buchanan",
    Address: {
      Street: "14 Garrett Hill",
      City: "London",
      Country: "UK",
    },
    Title: "Sales Manager",
    TitleOfCourtesy: "Mr.",
    BirthDate: ISODate("1955-03-04T00:00:00.000Z"),
    HireDate: ISODate("1993-10-17T00:00:00.000Z"),
    Phone: ["(71) 555-4848"],
    Salary: 2000,
  },
  {
    EmployeeID: 6,
    FirstName: "Michael",
    LastName: "Suyama",
    Address: {
      Street: "Coventry House Miner Rd.",
      City: "London",
      Country: "UK",
    },
    Title: "Sales Representative",
    TitleOfCourtesy: "Mr.",
    BirthDate: ISODate("1963-07-02T00:00:00.000Z"),
    HireDate: ISODate("1993-10-17T00:00:00.000Z"),
    Phone: ["(71) 555-7773"],
    Salary: 1500,
  },
],
```



```
{
  EmployeeID: 7,
  FirstName: "Robert",
  LastName: "King",
  Address: {
    Street: "Edgeham Hollow Winchester Way",
    City: "London",
    Country: "UK",
  },
  Title: "Sales Representative",
  TitleOfCourtesy: "Mr.",
  BirthDate: ISODate("1960-05-29T00:00:00.000Z"),
  HireDate: ISODate("1994-01-02T00:00:00.000Z"),
  Phone: ["(71) 555-5598"],
  Salary: 1000,
},
{
  EmployeeID: 8,
  FirstName: "Laura",
  LastName: "Callahan",
  Address: {
    Street: "4726 - 11th Ave. N.E.",
    City: "Seattle",
    Country: "USA",
  },
  Title: "Inside Sales Coordinator",
  TitleOfCourtesy: "Ms.",
  BirthDate: ISODate("1958-01-09T00:00:00.000Z"),
  HireDate: ISODate("1994-03-05T00:00:00.000Z"),
  Phone: ["(206) 555-1189"],
  Salary: 3000,
},
{
  EmployeeID: 9,
  FirstName: "Anne",
  LastName: "Dodsworth",
  Address: {
    Street: "7 Houndstooth Rd.",
    City: "London",
    Country: "UK",
  },
  Title: "Sales Representative",
  TitleOfCourtesy: "Ms.",
  BirthDate: ISODate("1966-01-27T00:00:00.000Z"),
  HireDate: ISODate("1994-11-15T00:00:00.000Z"),
  Phone: ["(71) 555-4444"],
  Salary: 1400,
},
];
```

- w **DataGrip** możesz posłużyć się poleceniami **SQL**
 - podzbiór poleceń **SQL** - proste polecenia **SELECT**
- wykonaj kilka takich poleceń
 - przeanalizuj wyniki
- np.

```
select * from employees;
select * from employees where Address.Country = "USA";
select * from employees where Address.Country = "USA" and TitleOfCourtesy = "Ms.";

select * from employees where Salary > 1200;
```

- napisz odpowiadające im polecenia **find**
- np

```
db.employees.find({});
db.employees.find({ "Address.Country": "USA" });
db.employees.find({
  $and: [{ "Address.Country": "USA" }, { TitleOfCourtesy: "Ms." }],
});

db.employees.find({
  $and: [{ "Address.Country": "USA" }, { TitleOfCourtesy: "Ms." }],
});

db.employees.find({ Salary: { $gt: 1200 } });
```

- operacja projekcji (wybór atrybutów w zbiorze wynikowym)

```
db.employees.find({});
```

```
db.employees.find({}, { FirstName: 1, LastName: 1 });
```

• wynik

```
[
  {
    _id: { $oid: "67e879dbb1c69e3d7e42c9c3" },
    FirstName: "Nancy",
    LastName: "Davolio",
  },
  {
    _id: { $oid: "67e879e2b1c69e3d7e42c9c5" },
    FirstName: "Andrew",
    LastName: "Fuller",
  },
  {
    _id: { $oid: "67e879e2b1c69e3d7e42c9c6" },
    FirstName: "Janet",
    LastName: "Leverling",
  },
  {
    _id: { $oid: "67e879e2b1c69e3d7e42c9c7" },
    FirstName: "Margaret",
    LastName: "Peacock",
  },
  {
    _id: { $oid: "67e879e2b1c69e3d7e42c9c8" },
    FirstName: "Steven",
    LastName: "Buchanan",
  },
  {
    _id: { $oid: "67e879e2b1c69e3d7e42c9c9" },
    FirstName: "Michael",
    LastName: "Suyama",
  },
  {
    _id: { $oid: "67e879e2b1c69e3d7e42c9ca" },
    FirstName: "Robert",
    LastName: "King",
  },
  {
    _id: { $oid: "67e879e2b1c69e3d7e42c9cb" },
    FirstName: "Laura",
    LastName: "Callahan",
  },
  {
    _id: { $oid: "67e879e2b1c69e3d7e42c9cc" },
    FirstName: "Anne",
    LastName: "Dodsworth",
  },
];
```

• warunek i projekcja

```
db.employees.find(
  {
    "Address.Country": "USA",
  },
  {
    FirstName: 1,
    LastName: 1,
  },
);
```

• wynik

```
[
  {
    FirstName: "Nancy",
    LastName: "Davolio",
  },
  {
    FirstName: "Andrew",
    LastName: "Fuller",
  },
  {
    FirstName: "Janet",
    LastName: "Leverling",
  },
];
```

```

    },
    {
      FirstName: "Margaret",
      LastName: "Peacock",
    },
    {
      FirstName: "Laura",
      LastName: "Callahan",
    },
  ],
];

```

- inne przykłady
 - 1 - atrybut pojawi się w zbiorze wynikowym
 - 0 - atrybut nie pojawi się
 - mogą wystąpić albo 1 albo 0
 - nie dotyczy atrybutu `_id`

```

db.employees.find({}, { _id: 0, FirstName: 1, LastName: 1 });

db.employees.find({}, { _id: 1, FirstName: 0, LastName: 0 });

// to jest błąd
db.employees.find({}, { _id: 1, FirstName: 1, LastName: 0 });

```

- `count`

```

select count(*) from employees;
select count(*) from employees where Address.Country = "USA";
select count(*) from employees where Address.Country = "USA" and TitleOfCourtesy = "Ms.";

select count(*) from employees where Salary > 1200;

```

```

db.employees.find({}).count();
db.employees.find({ "Address.Country": "USA" }).count();

db.employees
  .find({ $and: [{ "Address.Country": "USA" }, { TitleOfCourtesy: "Ms." }] })
  .count();

db.employees
  .find({
    $and: [{ "Address.Country": "USA" }, { TitleOfCourtesy: "Ms." }],
  })
  .count();

```

```

db.employees.find({ Salary: { $gt: 1200 } }).count();

```

- `wynik`

```

[
  {
    result: 5,
  },
];

```

- `$or $and`

```

select * from employees where Salary > 1100 and Salary < 1500

```

```

db.employees.find({
  Salary: { $gt: 1100, $lt: 1500 },
});

db.employees.find({
  $and: [{ Salary: { $gt: 1100 } }, { Salary: { $lt: 1500 } }],
});

```

- `wynik`

```

[
  {
    _id: { $oid: "67e879e2b1c69e3d7e42c9c6" },

```

```

Address: {
  Street: "722 Moss Bay Blvd.",
  City: "Kirkland",
  Country: "USA",
},
BirthDate: { $date: "1963-08-30T00:00:00.000Z" },
EmployeeID: 3,
FirstName: "Janet",
HireDate: { $date: "1992-04-01T00:00:00.000Z" },
LastName: "Leverling",
Phone: ["(206) 555-3412"],
Salary: 1200,
Title: "Sales Representative",
TitleOfCourtesy: "Ms.",
},
{
  _id: { $oid: "67e879e2b1c69e3d7e42c9cc" },
  Address: {
    Street: "7 Houndstooth Rd.",
    City: "London",
    Country: "UK",
  },
  BirthDate: { $date: "1966-01-27T00:00:00.000Z" },
  EmployeeID: 9,
  FirstName: "Anne",
  HireDate: { $date: "1994-11-15T00:00:00.000Z" },
  LastName: "Dodsworth",
  Phone: ["(71) 555-4444"],
  Salary: 1400,
  Title: "Sales Representative",
  TitleOfCourtesy: "Ms.",
},
];

```

```
select * from employees where Salary <= 1100 or Salary >= 1500
```

```

db.employees.find({
  $or: [{ Salary: { $lte: 1100 } }, { Salary: { $gte: 1500 } }],
});

```

- wynik

```

[
  {
    _id: { $oid: "67e879dbb1c69e3d7e42c9c3" },
    Address: {
      Street: "507 - 20th Ave. E.\r\nApt. 2A",
      City: "Seattle",
      Country: "USA",
    },
    BirthDate: { $date: "1948-12-08T00:00:00.000Z" },
    EmployeeID: 1,
    FirstName: "Nancy",
    HireDate: { $date: "1992-05-01T00:00:00.000Z" },
    LastName: "Davolio",
    Phone: ["(206) 555-9857", "(206) 555-9858"],
    Salary: 1000,
    Title: "Sales Representative",
    TitleOfCourtesy: "Ms.",
  },
  {
    _id: { $oid: "67e879e2b1c69e3d7e42c9c5" },
    Address: {
      Street: "908 W. Capital Way",
      City: "Tacoma",
      Country: "USA",
    },
    BirthDate: { $date: "1952-02-19T00:00:00.000Z" },
    EmployeeID: 2,
    FirstName: "Andrew",
    HireDate: { $date: "1992-08-14T00:00:00.000Z" },
    LastName: "Fuller",
    Phone: ["(206) 555-9482"],
    Salary: 10000,
    Title: "Vice President, Sales",
    TitleOfCourtesy: "Dr.",
  },
  {
    _id: { $oid: "67e879e2b1c69e3d7e42c9c7" },
    Address: {
      Street: "4110 Old Redmond Rd.",
      City: "Redmond",
      Country: "USA",
    },

```

```

    },
    BirthDate: { $date: "1937-09-19T00:00:00.000Z" },
    EmployeeID: 4,
    FirstName: "Margaret",
    HireDate: { $date: "1993-05-03T00:00:00.000Z" },
    LastName: "Peacock",
    Phone: ["(206) 555-8122"],
    Salary: 1100,
    Title: "Sales Representative",
    TitleOfCourtesy: "Mrs.",
  },
  {
    _id: { $oid: "67e879e2b1c69e3d7e42c9c8" },
    Address: {
      Street: "14 Garrett Hill",
      City: "London",
      Country: "UK",
    },
    BirthDate: { $date: "1955-03-04T00:00:00.000Z" },
    EmployeeID: 5,
    FirstName: "Steven",
    HireDate: { $date: "1993-10-17T00:00:00.000Z" },
    LastName: "Buchanan",
    Phone: ["(71) 555-4848"],
    Salary: 2000,
    Title: "Sales Manager",
    TitleOfCourtesy: "Mr.",
  },
  {
    _id: { $oid: "67e879e2b1c69e3d7e42c9c9" },
    Address: {
      Street: "Coventry House Miner Rd.",
      City: "London",
      Country: "UK",
    },
    BirthDate: { $date: "1963-07-02T00:00:00.000Z" },
    EmployeeID: 6,
    FirstName: "Michael",
    HireDate: { $date: "1993-10-17T00:00:00.000Z" },
    LastName: "Suyama",
    Phone: ["(71) 555-7773"],
    Salary: 1500,
    Title: "Sales Representative",
    TitleOfCourtesy: "Mr.",
  },
  {
    _id: { $oid: "67e879e2b1c69e3d7e42c9ca" },
    Address: {
      Street: "Edgeham Hollow Winchester Way",
      City: "London",
      Country: "UK",
    },
    BirthDate: { $date: "1960-05-29T00:00:00.000Z" },
    EmployeeID: 7,
    FirstName: "Robert",
    HireDate: { $date: "1994-01-02T00:00:00.000Z" },
    LastName: "King",
    Phone: ["(71) 555-5598"],
    Salary: 1000,
    Title: "Sales Representative",
    TitleOfCourtesy: "Mr.",
  },
  {
    _id: { $oid: "67e879e2b1c69e3d7e42c9cb" },
    Address: {
      Street: "4726 - 11th Ave. N.E.",
      City: "Seattle",
      Country: "USA",
    },
    BirthDate: { $date: "1958-01-09T00:00:00.000Z" },
    EmployeeID: 8,
    FirstName: "Laura",
    HireDate: { $date: "1994-03-05T00:00:00.000Z" },
    LastName: "Callahan",
    Phone: ["(206) 555-1189"],
    Salary: 3000,
    Title: "Inside Sales Coordinator",
    TitleOfCourtesy: "Ms.",
  },
];

```

```
select * from employees where year(HireDate) = 1992
```

```
db.employees.find({
  $expr: {
```

```
    $eq: [{ $year: "$HireDate" }, 1992],
  },
});

db.employees.find({
  HireDate: {
    $gte: ISODate("1992-01-01T00:00:00Z"),
    $lt: ISODate("1993-01-01T00:00:00Z"),
  },
});
```

- wynik

```
[
  {
    _id: { $oid: "67e879dbb1c69e3d7e42c9c3" },
    Address: {
      Street: "507 - 20th Ave. E.\r\nApt. 2A",
      City: "Seattle",
      Country: "USA",
    },
    BirthDate: { $date: "1948-12-08T00:00:00.000Z" },
    EmployeeID: 1,
    FirstName: "Nancy",
    HireDate: { $date: "1992-05-01T00:00:00.000Z" },
    LastName: "Davolio",
    Phone: ["(206) 555-9857", "(206) 555-9858"],
    Salary: 1000,
    Title: "Sales Representative",
    TitleOfCourtesy: "Ms.",
  },
  {
    _id: { $oid: "67e879e2b1c69e3d7e42c9c5" },
    Address: {
      Street: "908 W. Capital Way",
      City: "Tacoma",
      Country: "USA",
    },
    BirthDate: { $date: "1952-02-19T00:00:00.000Z" },
    EmployeeID: 2,
    FirstName: "Andrew",
    HireDate: { $date: "1992-08-14T00:00:00.000Z" },
    LastName: "Fuller",
    Phone: ["(206) 555-9482"],
    Salary: 10000,
    Title: "Vice President, Sales",
    TitleOfCourtesy: "Dr.",
  },
  {
    _id: { $oid: "67e879e2b1c69e3d7e42c9c6" },
    Address: {
      Street: "722 Moss Bay Blvd.",
      City: "Kirkland",
      Country: "USA",
    },
    BirthDate: { $date: "1963-08-30T00:00:00.000Z" },
    EmployeeID: 3,
    FirstName: "Janet",
    HireDate: { $date: "1992-04-01T00:00:00.000Z" },
    LastName: "Leverling",
    Phone: ["(206) 555-3412"],
    Salary: 1200,
    Title: "Sales Representative",
    TitleOfCourtesy: "Ms.",
  },
];
```

Zadanie 2

- Wykonaj kilka własnych eksperymentów

Zadanie 2 - rozwiązanie

Wyniki:

przykłady, kod, zrzuty ekranów, komentarz ...

```
use employees;
db.employees.insertMany([
  {
    EmployeeID: 1,
    FirstName: "Nancy",
    LastName: "Davolio",
    Address: {
```

```

    Street: "507 - 20th Ave. E. Apt. 2A",
    City: "Seattle",
    Country: "USA",
  },
  Title: "Sales Representative",
  TitleOfCourtesy: "Ms.",
  BirthDate: ISODate("1948-12-08T00:00:00.000Z"),
  HireDate: ISODate("1992-05-01T00:00:00.000Z"),
  Phone: ["(206) 555-9857", "(206) 555-9858"],
  Salary: 1000,
},
{
  EmployeeID: 2,
  FirstName: "Andrew",
  LastName: "Fuller",
  Address: {
    Street: "908 W. Capital Way",
    City: "Tacoma",
    Country: "USA",
  },
  Title: "Vice President, Sales",
  TitleOfCourtesy: "Dr.",
  BirthDate: ISODate("1952-02-19T00:00:00.000Z"),
  HireDate: ISODate("1992-08-14T00:00:00.000Z"),
  Phone: ["(206) 555-9482"],
  Salary: 10000,
},
{
  EmployeeID: 3,
  FirstName: "Janet",
  LastName: "Leverling",
  Address: {
    Street: "722 Moss Bay Blvd.",
    City: "Kirkland",
    Country: "USA",
  },
  Title: "Sales Representative",
  TitleOfCourtesy: "Ms.",
  BirthDate: ISODate("1963-08-30T00:00:00.000Z"),
  HireDate: ISODate("1992-04-01T00:00:00.000Z"),
  Phone: ["(206) 555-3412"],
  Salary: 1200,
},
{
  EmployeeID: 4,
  FirstName: "Margaret",
  LastName: "Peacock",
  Address: {
    Street: "4110 Old Redmond Rd.",
    City: "Redmond",
    Country: "USA",
  },
  Title: "Sales Representative",
  TitleOfCourtesy: "Mrs.",
  BirthDate: ISODate("1937-09-19T00:00:00.000Z"),
  HireDate: ISODate("1993-05-03T00:00:00.000Z"),
  Phone: ["(206) 555-8122"],
  Salary: 1100,
},
{
  EmployeeID: 5,
  FirstName: "Steven",
  LastName: "Buchanan",
  Address: {
    Street: "14 Garrett Hill",
    City: "London",
    Country: "UK",
  },
  Title: "Sales Manager",
  TitleOfCourtesy: "Mr.",
  BirthDate: ISODate("1955-03-04T00:00:00.000Z"),
  HireDate: ISODate("1993-10-17T00:00:00.000Z"),
  Phone: ["(71) 555-4848"],
  Salary: 2000,
},
{
  EmployeeID: 6,
  FirstName: "Michael",
  LastName: "Suyama",
  Address: {
    Street: "Coventry House Miner Rd.",
    City: "London",
    Country: "UK",
  },
  Title: "Sales Representative",
  TitleOfCourtesy: "Mr.",
  BirthDate: ISODate("1963-07-02T00:00:00.000Z"),
  HireDate: ISODate("1993-10-17T00:00:00.000Z"),
  Phone: ["(71) 555-7773"],
  Salary: 1500,
}

```

```

    },
    {
      EmployeeID: 7,
      FirstName: "Robert",
      LastName: "King",
      Address: {
        Street: "Edgeham Hollow Winchester Way",
        City: "London",
        Country: "UK",
      },
      Title: "Sales Representative",
      TitleOfCourtesy: "Mr.",
      BirthDate: ISODate("1960-05-29T00:00:00.000Z"),
      HireDate: ISODate("1994-01-02T00:00:00.000Z"),
      Phone: ["(71) 555-5598"],
      Salary: 1000,
    },
    {
      EmployeeID: 8,
      FirstName: "Laura",
      LastName: "Callahan",
      Address: {
        Street: "4726 - 11th Ave. N.E.",
        City: "Seattle",
        Country: "USA",
      },
      Title: "Inside Sales Coordinator",
      TitleOfCourtesy: "Ms.",
      BirthDate: ISODate("1958-01-09T00:00:00.000Z"),
      HireDate: ISODate("1994-03-05T00:00:00.000Z"),
      Phone: ["(206) 555-1189"],
      Salary: 3000,
    },
    {
      EmployeeID: 9,
      FirstName: "Anne",
      LastName: "Dodsworth",
      Address: {
        Street: "7 Houndstooth Rd.",
        City: "London",
        Country: "UK",
      },
      Title: "Sales Representative",
      TitleOfCourtesy: "Ms.",
      BirthDate: ISODate("1966-01-27T00:00:00.000Z"),
      HireDate: ISODate("1994-11-15T00:00:00.000Z"),
      Phone: ["(71) 555-4444"],
      Salary: 1400,
    },
  ]});

// Pracownicy, którzy mają dwa numery telefonu
db.employees.find({ Phone: { $size: 2 } });

// Pracownicy o imieniu Steven
db.employees.find({FirstName: "Steven"})

// Pracownicy na 'A'
db.employees.find({FirstName: /^A/})

// Pracownicy z Wielkiej Brytanii, zarabiający więcej niż 1200
db.employees.find({"Address.Country": "UK", Salary: {$gt: 1200}})
select * from employees where Address.Country = "UK" and Salary > 1200

// Pracownicy zatrudnieni po 1992, z UK, albo mający w nazwie stanowiska "Representative" albo zarabiający więcej niż 1500
db.employees.find({
  "Address.Country": "UK",
  HireDate: { $gt: ISODate("1992-01-01T00:00:00.000Z") }
  $or: [
    { Title: /Representative/ },
    { Salary: { $gt: 1500} }
  ],
})

```

Operacje agregacji

- aggregate
- aggregation pipeline
- <https://www.mongodb.com/docs/manual/core/aggregation-pipeline/>

```

db.<collection>.aggregate(

[

```



```

    {stage 1},

    {stage 2},

    ...

    {stage N}

]

)

```

- <https://www.mongodb.com/docs/manual/reference/operator/aggregation-pipeline/#std-label-aggregation-pipeline-operator-reference>
- Stages
 - \$match
 - \$project
 - \$group
 - \$unwind
 - \$lookup
 - \$out
 - ...
- Aggregation vs SQL
 - <https://www.mongodb.com/docs/manual/reference/sql-aggregation-comparison/>

Przykład 3

- baza `employees` z poprzedniego przykładu

```

use employees;

db;

db.employees.find();

```

- Operacje agregacji
 - `aggregate`

```

db.employees.aggregate([
  {
    $match: {
      "Address.Country": "USA",
    },
  },
  {
    $project: {
      _id: 0,
      LastName: 1,
      FirstName: 1,
    },
  },
  {
    $sort: { LastName: -1 },
  },
]);

```

- wynik

```

[
  {
    FirstName: "Nancy",
    LastName: "Davolio",
  },
  {
    FirstName: "Andrew",
    LastName: "Fuller",
  },
  {
    FirstName: "Janet",
    LastName: "Leverling",
  },
  {
    FirstName: "Margaret",
    LastName: "Peacock",
  },
  {

```

```

    FirstName: "Laura",
    LastName: "Callahan",
  },
];

```

- dodatkowe pole `FullName`

```

db.employees.aggregate([
  {
    $match: {
      "Address.Country": "USA",
    },
  },
  {
    $project: {
      _id: 0,
      First: "$LastName",
      Last: "$LastName",
      FullName: {
        $concat: ["$FirstName", " ", "$LastName"],
      },
    },
  },
  {
    $sort: { lastname: -1 },
  },
]);

```

- wynik

```

[
  {
    First: "Davolio",
    FullName: "Nancy Davolio",
    Last: "Davolio",
  },
  {
    First: "Fuller",
    FullName: "Andrew Fuller",
    Last: "Fuller",
  },
  {
    First: "Leverling",
    FullName: "Janet Leverling",
    Last: "Leverling",
  },
  {
    First: "Peacock",
    FullName: "Margaret Peacock",
    Last: "Peacock",
  },
  {
    First: "Callahan",
    FullName: "Laura Callahan",
    Last: "Callahan",
  },
];

```

- grupowanie
 - `$group`

```

select TitleOfCourtesy, count(*)
from employees
where Address.Country = "USA"
group by TitleOfCourtesy

```

```

db.employees.aggregate([
  {
    $match: {
      "Address.Country": "USA",
    },
  },
  {
    $group: {
      _id: "$TitleOfCourtesy",
      cnt: { $sum: 1 },
    },
  },
]);

```

- wynik

```
[
  {
    _id: "Dr.",
    cnt: 1,
  },
  {
    _id: "Mrs.",
    cnt: 1,
  },
  {
    _id: "Ms.",
    cnt: 3,
  },
];
```

- dodatkowa projekcja, zmiana nazwy

```
db.employees.aggregate([
  {
    $match: {
      "Address.Country": "USA",
    },
  },
  {
    $group: {
      _id: "$TitleOfCourtesy",
      cnt: { $sum: 1 },
    },
  },
  {
    $project: {
      _id: 0,
      TitleOfCourtesy: "$_id",
      cnt: 1,
    },
  },
]);
```

- grupowanie wg kilku atrybutów

```
select Address.Country, TitleOfCourtesy, count(*) cnt
from employees
group by Address.Country, TitleOfCourtesy
order by cnt desc;
```

```
db.employees.aggregate([
  {
    $group: {
      _id: {
        country: "$Address.Country",
        title: "$TitleOfCourtesy",
      },
      cnt: { $sum: 1 },
    },
  },
  {
    $project: {
      _id: 0,
      country: "$_id.country",
      title: "$_id.title",
      cnt: 1,
    },
  },
  {
    $sort: { cnt: -1 },
  },
]);
```

- wynik

```
[
  {
    Country: "UK",
    TitleOfCourtesy: "Mr.",
    cnt: 3,
  },
  {
    Country: "USA",
```

```

    TitleOfCourtesy: "Ms.",
    cnt: 3,
  },
  {
    Country: "USA",
    TitleOfCourtesy: "Dr.",
    cnt: 1,
  },
  {
    Country: "UK",
    TitleOfCourtesy: "Ms.",
    cnt: 1,
  },
  {
    Country: "USA",
    TitleOfCourtesy: "Mrs.",
    cnt: 1,
  },
];

```

- zapis wyniku
 - `$out`
 - zapis zbioru wynikowego to kolekcji o wskazanej nazwie

```

db.employees.aggregate([
  {
    $group: {
      _id: {
        Country: "$Address.Country",
        TitleOfCourtesy: "$TitleOfCourtesy",
      },
      cnt: { $sum: 1 },
    },
  },
  {
    $project: {
      _id: 0,
      Country: "$_id.Country",
      TitleOfCourtesy: "$_id.TitleOfCourtesy",
      cnt: 1,
    },
  },
  {
    $out: "em_by_country_title",
  },
]);

```

- lista kolekcji

```
show collections;
```

- wynik

```

[
  {
    badge: "",
    name: "em_by_country_title",
  },
  {
    badge: "",
    name: "employees",
  },
];

```

- zawartość kolekcji `em_by_cuntry_title`

```
db.em_by_country_title.find();
```

- wynik

```

[
  {
    _id: { $oid: "67ec20f77b1ef941c9512dcb" },
    Country: "USA",
    TitleOfCourtesy: "Mrs.",
    cnt: 1,
  },
  {
    _id: { $oid: "67ec20f77b1ef941c9512dcc" },

```

```
Country: "USA",
TitleOfCourtesy: "Ms.",
cnt: 3,
},
{
  _id: { $oid: "67ec20f77b1ef941c9512dcd" },
Country: "USA",
TitleOfCourtesy: "Dr.",
cnt: 1,
},
{
  _id: { $oid: "67ec20f77b1ef941c9512dce" },
Country: "UK",
TitleOfCourtesy: "Mr.",
cnt: 3,
},
{
  _id: { $oid: "67ec20f77b1ef941c9512dcf" },
Country: "UK",
TitleOfCourtesy: "Ms.",
cnt: 1,
},
];
```

Zadanie 3

- Wykonaj kilka własnych eksperymentów

Zadanie 3 - rozwiązanie

Wyniki:

przykłady, kod, zrzuty ekranów, komentarz ...

```
// Średnie zarabki pracowników w poszczególnych krajach
db.employees.aggregate([
  {
    $group: {
      _id: "$Address.Country",
      "Average salary": { $avg: "$Salary" },
    },
  },
  {
    $project: {
      _id: 0,
      Country: "$_id",
      "Average salary": { $round: ["$Average salary", 2] },
    },
  },
  {
    $sort: { "Average salary": -1 },
  },
]);

// Największa pensja z podziałem na stanowisko
db.employees.aggregate([
  {
    $group: {
      _id: "$Title",
      "Highest salary": { $max: "$Salary" },
    },
  },
  {
    $project: {
      _id: 0,
      Title: "$_id",
      "Highest salary": { $round: ["$Highest salary", 2] },
    },
  },
]);

// Liczba pracowników urodzona w danym roku
db.employees.aggregate([
  {
    $group: {
      _id: { $year: "$BirthDate" },
      "Employee counter": { $sum: 1 },
    },
  },
  {
    $project: {
      _id: 0,
      "Year of birth": "$_id",
      "Employee counter": 1,
    },
  },
]);
```

```
},  
});
```

Przykład 4

Odtwórz z backupu bazę `north0`

- plik z z backup'em bazy danych dostępny jest w moodle
- najprostsza wersja
 - `mongorestore <folder>`

```
mongorestore dump
```

- to polecenie odtworzy wszystkie bazy danych znajdujące się we wskazanym folderze (w tym przypadku `dump`)
 - najłatwiej wgrać tam folder zawierający pliki z backupem i wykonać proste polecenie `mongorestore`
- dokumentacja:
 - <https://www.mongodb.com/docs/database-tools/mongorestore/>

Wybierz bazę `north0`

```
use north0
```

Baza `north0` jest kopią relacyjnej bazy danych `Northwind`

- poszczególne kolekcje odpowiadają tabelom w oryginalnej bazie `Northwind`

zapoznaj się ze strukturą dokumentów w bazie `North0`

```
db.customers.find();  
db.orders.find();  
db.orderdetails.find();
```

Przykład 5

produkty, kategorie, dostawcy

```
db.categories.find().limit(2);
```

- wynik

```
[  
  {  
    _id: { $oid: "67eb06a27b1ef941c950fb49" },  
    CategoryID: 1,  
    CategoryName: "Beverages",  
    Description: "Soft drinks, coffees, teas, beers, and ales",  
  },  
  {  
    _id: { $oid: "67eb06a27b1ef941c950fb4a" },  
    CategoryID: 2,  
    CategoryName: "Condiments",  
    Description: "Sweet and savory sauces, relishes, spreads, and seasonings",  
  },  
];
```

```
db.products.find().limit(2);
```

- wynik

```
[  
  {  
    _id: { $oid: "63a0607abb3b972d6f4e1f50" },  
    CategoryID: 1,  
    Discontinued: false,  
    ProductID: 1,  
    ProductName: "Chai",  
    QuantityPerUnit: "10 boxes x 20 bags",  
    ReorderLevel: 10,  
  },  
];
```

```

    SupplierID: 1,
    UnitPrice: 18,
    UnitsInStock: 39,
    UnitsOnOrder: 0,
  },
  {
    _id: { $oid: "63a0607abb3b972d6f4e1f51" },
    CategoryID: 1,
    Discontinued: false,
    ProductID: 2,
    ProductName: "Chang",
    QuantityPerUnit: "24 - 12 oz bottles",
    ReorderLevel: 25,
    SupplierID: 1,
    UnitPrice: 19,
    UnitsInStock: 17,
    UnitsOnOrder: 40,
  },
];

```

```
db.suppliers.find().limit(2);
```

- wynik

```

[
  {
    _id: { $oid: "63a05d3cbb3b972d6f4e09f7" },
    Address: "49 Gilbert St.",
    City: "London",
    CompanyName: "Exotic Liquids",
    ContactName: "Charlotte Cooper",
    ContactTitle: "Purchasing Manager",
    Country: "UK",
    Fax: null,
    HomePage: null,
    Phone: "(171) 555-2222",
    PostalCode: "EC1 4SD",
    Region: null,
    SupplierID: 1,
  },
  {
    _id: { $oid: "63a05d3cbb3b972d6f4e09f8" },
    Address: "P.O. Box 78934",
    City: "New Orleans",
    CompanyName: "New Orleans Cajun Delights",
    ContactName: "Shelley Burke",
    ContactTitle: "Order Administrator",
    Country: "USA",
    Fax: null,
    HomePage: "#CAJUN.HTM#",
    Phone: "(100) 555-4822",
    PostalCode: "70117",
    Region: "LA",
    SupplierID: 2,
  },
];

```

Uwaga:

- etap `$match` z pustym warunkiem zwraca wszystkie dokumenty (nie jest potrzebny, można go pominąć)
 - w przykładzie etap został dodany żeby łatwiej można było wykonać własne eksperymenty ograniczające zbiór dokumentów z kolekcji `products`

```

db.products.aggregate([
  {
    $match: {},
  },
  //suppliers
  {
    $lookup: {
      from: "suppliers",
      localField: "SupplierID",
      foreignField: "SupplierID",
      as: "Supplier",
    },
  },
  {
    $limit: 2,
  },
]);

```

- pole `Supplier` jest tablicą

- zawiera 1 obiekt

- wynik

```
[
  {
    _id: { $oid: "63a0607abb3b972d6f4e1f50" },
    CategoryID: 1,
    Discontinued: false,
    ProductID: 1,
    ProductName: "Chai",
    QuantityPerUnit: "10 boxes x 20 bags",
    ReorderLevel: 10,
    Supplier: [
      {
        _id: { $oid: "63a05d3cbb3b972d6f4e09f7" },
        SupplierID: 1,
        CompanyName: "Exotic Liquids",
        ContactName: "Charlotte Cooper",
        ContactTitle: "Purchasing Manager",
        Address: "49 Gilbert St.",
        City: "London",
        Region: null,
        PostalCode: "EC1 4SD",
        Country: "UK",
        Phone: "(171) 555-2222",
        Fax: null,
        HomePage: null,
      },
    ],
    SupplierID: 1,
    UnitPrice: 18,
    UnitsInStock: 39,
    UnitsOnOrder: 0,
  },
  {
    _id: { $oid: "63a0607abb3b972d6f4e1f51" },
    CategoryID: 1,
    Discontinued: false,
    ProductID: 2,
    ProductName: "Chang",
    QuantityPerUnit: "24 - 12 oz bottles",
    ReorderLevel: 25,
    Supplier: [
      {
        _id: { $oid: "63a05d3cbb3b972d6f4e09f7" },
        SupplierID: 1,
        CompanyName: "Exotic Liquids",
        ContactName: "Charlotte Cooper",
        ContactTitle: "Purchasing Manager",
        Address: "49 Gilbert St.",
        City: "London",
        Region: null,
        PostalCode: "EC1 4SD",
        Country: "UK",
        Phone: "(171) 555-2222",
        Fax: null,
        HomePage: null,
      },
    ],
    SupplierID: 1,
    UnitPrice: 19,
    UnitsInStock: 17,
    UnitsOnOrder: 40,
  },
];
```

- \$unwind
 - pole `Supplier` jest obiektem/dokumentem

```
db.products.aggregate([
  {
    $match: {},
  },

  //suppliers
  {
    $lookup: {
      from: "suppliers",
      localField: "SupplierID",
      foreignField: "SupplierID",
      as: "Supplier",
    },
  },
  {
    $unwind: "$Supplier",
```



```

    },
    {
      $limit: 2,
    },
  ],
});

```

- wynik

```

[
  {
    _id: { $oid: "63a0607abb3b972d6f4e1f50" },
    CategoryID: 1,
    Discontinued: false,
    ProductID: 1,
    ProductName: "Chai",
    QuantityPerUnit: "10 boxes x 20 bags",
    ReorderLevel: 10,
    Supplier: {
      _id: { $oid: "63a05d3cbb3b972d6f4e09f7" },
      SupplierID: 1,
      CompanyName: "Exotic Liquids",
      ContactName: "Charlotte Cooper",
      ContactTitle: "Purchasing Manager",
      Address: "49 Gilbert St.",
      City: "London",
      Region: null,
      PostalCode: "EC1 4SD",
      Country: "UK",
      Phone: "(171) 555-2222",
      Fax: null,
      HomePage: null,
    },
    SupplierID: 1,
    UnitPrice: 18,
    UnitsInStock: 39,
    UnitsOnOrder: 0,
  },
  {
    _id: { $oid: "63a0607abb3b972d6f4e1f51" },
    CategoryID: 1,
    Discontinued: false,
    ProductID: 2,
    ProductName: "Chang",
    QuantityPerUnit: "24 - 12 oz bottles",
    ReorderLevel: 25,
    Supplier: {
      _id: { $oid: "63a05d3cbb3b972d6f4e09f7" },
      SupplierID: 1,
      CompanyName: "Exotic Liquids",
      ContactName: "Charlotte Cooper",
      ContactTitle: "Purchasing Manager",
      Address: "49 Gilbert St.",
      City: "London",
      Region: null,
      PostalCode: "EC1 4SD",
      Country: "UK",
      Phone: "(171) 555-2222",
      Fax: null,
      HomePage: null,
    },
    SupplierID: 1,
    UnitPrice: 19,
    UnitsInStock: 17,
    UnitsOnOrder: 40,
  },
];

```

- projekcja
 - tylko wybrane pola z `Supplier`

```

db.products.aggregate([
  {
    $match: {},
  },

  //suppliers
  {
    $lookup: {
      from: "suppliers",
      localField: "SupplierID",
      foreignField: "SupplierID",
      as: "Supplier_tmp",
    },
  },
  {

```

```

    $unwind: "$Supplier_tmp",
  },
  {
    $addFields: {
      Supplier: {
        SupplierID: "$SupplierID",
        CompanyName: "$Supplier_tmp.CompanyName",
        Country: "$Supplier_tmp.Country",
      },
    },
  },
  {
    $project: {
      SupplierID: 0,
      Supplier_tmp: 0,
    },
  },
  {
    $limit: 2,
  },
});

```

- wynik

```

[
  {
    _id: { $oid: "63a0607abb3b972d6f4e1f50" },
    CategoryID: 1,
    Discontinued: false,
    ProductID: 1,
    ProductName: "Chai",
    QuantityPerUnit: "10 boxes x 20 bags",
    ReorderLevel: 10,
    Supplier: {
      SupplierID: 1,
      CompanyName: "Exotic Liquids",
      Country: "UK",
    },
    UnitPrice: 18,
    UnitsInStock: 39,
    UnitsOnOrder: 0,
  },
  {
    _id: { $oid: "63a0607abb3b972d6f4e1f51" },
    CategoryID: 1,
    Discontinued: false,
    ProductID: 2,
    ProductName: "Chang",
    QuantityPerUnit: "24 - 12 oz bottles",
    ReorderLevel: 25,
    Supplier: {
      SupplierID: 1,
      CompanyName: "Exotic Liquids",
      Country: "UK",
    },
    UnitPrice: 19,
    UnitsInStock: 17,
    UnitsOnOrder: 40,
  },
];

```

- dodaj do wyniku inf. o kategoriach

```

db.products.aggregate([
  {
    $match: {},
  },
  //suppliers
  {
    $lookup: {
      from: "suppliers",
      localField: "SupplierID",
      foreignField: "SupplierID",
      as: "Supplier_tmp",
    },
  },
  //categories
  {
    $lookup: {
      from: "categories",
      localField: "CategoryID",
      foreignField: "CategoryID",
      as: "Category_tmp",
    },
  },
]);

```

```

    {
      $unwind: "$Supplier_tmp",
    },
    {
      $unwind: "$Category_tmp",
    },
    {
      $addFields: {
        Supplier: {
          SupplierID: "$SupplierID",
          CompanyName: "$Supplier_tmp.CompanyName",
          Country: "$Supplier_tmp.Country",
        },
        Category: {
          CategoryID: "$CategoryID",
          CategorynameName: "$Category_tmp.CategoryName",
        },
      },
    },
  },
  {
    $project: {
      SupplierID: 0,
      Supplier_tmp: 0,
      CategoryID: 0,
      Category_tmp: 0,
    },
  },
  {
    $limit: 2,
  },
  {});

```

- wynik

```

[
  {
    _id: { $oid: "63a0607abb3b972d6f4e1f50" },
    Category: {
      CategoryID: 1,
      CategorynameName: "Beverages",
    },
    Discontinued: false,
    ProductID: 1,
    ProductName: "Chai",
    QuantityPerUnit: "10 boxes x 20 bags",
    ReorderLevel: 10,
    Supplier: {
      SupplierID: 1,
      CompanyName: "Exotic Liquids",
      Country: "UK",
    },
    UnitPrice: 18,
    UnitsInStock: 39,
    UnitsOnOrder: 0,
  },
  {
    _id: { $oid: "63a0607abb3b972d6f4e1f51" },
    Category: {
      CategoryID: 1,
      CategorynameName: "Beverages",
    },
    Discontinued: false,
    ProductID: 2,
    ProductName: "Chang",
    QuantityPerUnit: "24 - 12 oz bottles",
    ReorderLevel: 25,
    Supplier: {
      SupplierID: 1,
      CompanyName: "Exotic Liquids",
      Country: "UK",
    },
    UnitPrice: 19,
    UnitsInStock: 17,
    UnitsOnOrder: 40,
  },
];

```

Przykład 6

- dostawcy i produkty

```

db.suppliers.aggregate([
  {

```

```

    $match: {},
  },

  //products
  {
    $lookup: {
      from: "products",
      localField: "SupplierID",
      foreignField: "SupplierID",
      as: "Products",
    },
  },
  {
    $addFields: {
      Products: {
        $map: {
          input: "$Products",
          as: "product",
          in: {
            ProductID: "$$product.ProductID",
            ProductName: "$$product.ProductName",
            UnitPrice: "$$product.UnitPrice",
            Discontinued: "$$product.Discontinued",
          },
        },
      },
    },
  },
  {
    $limit: 2,
  },
}];

```

• wynik

```

[
  {
    _id: { $oid: "63a05d3cbb3b972d6f4e09f7" },
    Address: "49 Gilbert St.",
    City: "London",
    CompanyName: "Exotic Liquids",
    ContactName: "Charlotte Cooper",
    ContactTitle: "Purchasing Manager",
    Country: "UK",
    Fax: null,
    HomePage: null,
    Phone: "(171) 555-2222",
    PostalCode: "EC1 4SD",
    Products: [
      {
        ProductID: 1,
        ProductName: "Chai",
        UnitPrice: 18,
        Discontinued: false,
      },
      {
        ProductID: 2,
        ProductName: "Chang",
        UnitPrice: 19,
        Discontinued: false,
      },
      {
        ProductID: 3,
        ProductName: "Aniseed Syrup",
        UnitPrice: 10,
        Discontinued: false,
      },
    ],
    Region: null,
    SupplierID: 1,
  },
  {
    _id: { $oid: "63a05d3cbb3b972d6f4e09f8" },
    Address: "P.O. Box 78934",
    City: "New Orleans",
    CompanyName: "New Orleans Cajun Delights",
    ContactName: "Shelley Burke",
    ContactTitle: "Order Administrator",
    Country: "USA",
    Fax: null,
    HomePage: "#CAJUN.HTM#",
    Phone: "(100) 555-4822",
    PostalCode: "70117",
    Products: [
      {
        ProductID: 4,
        ProductName: "Chef Anton's Cajun Seasoning",
        UnitPrice: 22,

```

```

    Discontinued: false,
  },
  {
    ProductID: 5,
    ProductName: "Chef Anton's Gumbo Mix",
    UnitPrice: 21.35,
    Discontinued: true,
  },
  {
    ProductID: 65,
    ProductName: "Louisiana Fiery Hot Pepper Sauce",
    UnitPrice: 21.05,
    Discontinued: false,
  },
  {
    ProductID: 66,
    ProductName: "Louisiana Hot Spiced Okra",
    UnitPrice: 17,
    Discontinued: false,
  },
],
Region: "LA",
SupplierID: 2,
},
];

```

Przykład 7

- produkty
 - tworzymy kolekcję `products_tmp` (połączenie danych z kolekcji `products`, `suppliers`, `categories`)

```

db.products.aggregate([
  { $match: {} },
  //suppliers
  {
    $lookup: {
      from: "suppliers",
      localField: "SupplierID",
      foreignField: "SupplierID",
      as: "supplier",
    },
  },
  { $unwind: "$supplier" },
  {
    $project: {
      "supplier._id": 0,
      "supplier.ContactName": 0,
      "supplier.ContactTitle": 0,
      "supplier.City": 0,
      "supplier.Country": 0,
      "supplier.Address": 0,
      "supplier.PostalCode": 0,
      "supplier.Region": 0,
      "supplier.Phone": 0,
      "supplier.Fax": 0,
      "supplier.HomePage": 0,
    },
  },
  //categories
  {
    $lookup: {
      from: "categories",
      localField: "CategoryID",
      foreignField: "CategoryID",
      as: "category",
    },
  },
  { $unwind: "$category" },
  {
    $project: {
      "category._id": 0,
      "category.Description": 0,
      "category.Picture": 0,
    },
  },
  //usunięcie kluczy obcych
  {
    $project: {
      Discontinued: 0,
      QuantityPerUnit: 0,
      ReorderLevel: 0,
      UnitPrice: 0,
      UnitsInStock: 0,
      UnitsOnOrder: 0,
    },
  },
]);

```

```

    },
    {
      $project: {
        ProductID: 1,
        ProductName: 1,
        CategoryID: 1,
        CategoryName: "$category.CategoryName",
        SupplierID: 1,
        SupplierName: "$supplier.CompanyName",
      },
    },
    { $out: "products_tmp" },
  ]);

```

```
db.products_tmp.find();
```

- pozycje zamówień
 - tworzymy kolekcję `orderdetails_tmp` (polecenie wykorzystuje `products_tmp` stworzoną w poprzednim przykładzie)

```

db.orderdetails.aggregate(
  { $match: {} },

  //suppliers
  {
    $lookup: {
      from: "products_tmp",
      localField: "ProductID",
      foreignField: "ProductID",
      as: "product",
    },
  },
  { $unwind: "$product" },
  {
    $project: {
      "product._id": 0,
    },
  },
  {
    $project: {
      ProductID: 0,
    },
  },
  {
    $addFields: {
      TotalValue: {
        $multiply: ["$UnitPrice", "$Quantity", { $subtract: [1, "$Discount"] }],
      },
      Discount: { $round: ["$Discount", 2] },
    },
  },
  { $out: "orderdetails_tmp" },
);

```

```
db.orderdetails_tmp.find();
```

- zamówienia
 - każdy dokument agreguje pełną informację o zamówieniu
 - tworzymy kolekcję `orders_tmp` (polecenie wykorzystuje `orderdetails_tmp` stworzoną w poprzednim przykładzie)

```

db.orders.aggregate(
  {
    $match: {},
  },

  //customers
  {
    $lookup: {
      from: "customers",
      localField: "CustomerID",
      foreignField: "CustomerID",
      as: "Customer",
    },
  },
  {
    $unwind: "$Customer",
  },
  {
    $project: {
      "Customer._id": 0,
      "Customer.ContactName": 0,
    },
  },
);

```

```

    "Customer.ContactTitle": 0,
    "Customer.City": 0,
    // "Customer.Country": 0,
    "Customer.Address": 0,
    "Customer.PostalCode": 0,
    "Customer.Region": 0,
    "Customer.Phone": 0,
    "Customer.Fax": 0,
  },
},
//employees
{
  $lookup: {
    from: "employees",
    localField: "EmployeeID",
    foreignField: "EmployeeID",
    as: "Employee",
  },
},
{
  $unwind: "$Employee",
},
{
  $project: {
    "Employee._id": 0,
    "Employee.Title": 0,
    "Employee.TitleOfCourtesy": 0,
    "Employee.BirthDate": 0,
    "Employee.HireDate": 0,
    "Employee.Address": 0,
    "Employee.PostalCode": 0,
    "Employee.City": 0,
    "Employee.Region": 0,
    "Employee.Country": 0,
    "Employee.HomePhone": 0,
    "Employee.Extension": 0,
    "Employee.Photo": 0,
    "Employee.Notes": 0,
    "Employee.ReportsTo": 0,
    "Employee.PhotoPath": 0,
  },
},
//shippers
{
  $lookup: {
    from: "shippers",
    localField: "ShipVia",
    foreignField: "ShipperID",
    as: "Shipper",
  },
},
{
  $unwind: "$Shipper",
},
{
  $project: {
    "Shipper._id": 0,
    "Shipper.Phone": 0,
  },
},
//orderdetails
{
  $lookup: {
    from: "orderdetails_tmp",
    localField: "OrderID",
    foreignField: "OrderID",
    as: "Orderdetails",
  },
},
{
  $project: {
    "Orderdetails._id": 0,
    "Orderdetails.OrderID": 0,
  },
},
{
  $project: {
    CustomerID: 0,
    EmployeeID: 0,
    ShipVia: 0,
  },
},
{
  $out: "orders_tmp",
},
);

```

```
db.orders_tmp.find().limit(2);
```

• wynik

```
[
  {
    _id: { $oid: "63a060b9bb3b972d6f4e1fc6" },
    Customer: {
      CustomerID: "VINET",
      CompanyName: "Vins et alcools Chevalier",
      Country: "France",
    },
    Employee: {
      EmployeeID: 5,
      LastName: "Buchanan",
      FirstName: "Steven",
    },
    Freight: 32.38,
    OrderDate: { $date: "1996-07-04T00:00:00.000Z" },
    OrderID: 10248,
    Orderdetails: [
      {
        UnitPrice: 14,
        Quantity: 12,
        Discount: 0,
        product: {
          ProductID: 11,
          ProductName: "Queso Cabrales",
          QuantityPerUnit: "1 kg pkg.",
        },
        TotalValue: 168,
      },
      {
        UnitPrice: 9.8,
        Quantity: 10,
        Discount: 0,
        product: {
          ProductID: 42,
          ProductName: "Singaporean Hokkien Fried Mee",
          QuantityPerUnit: "32 - 1 kg pkgs.",
        },
        TotalValue: 98,
      },
      {
        UnitPrice: 34.8,
        Quantity: 5,
        Discount: 0,
        product: {
          ProductID: 72,
          ProductName: "Mozzarella di Giovanni",
          QuantityPerUnit: "24 - 200 g pkgs.",
        },
        TotalValue: 174,
      },
    ],
    RequiredDate: { $date: "1996-08-01T00:00:00.000Z" },
    ShipAddress: "59 rue de l'Abbaye",
    ShipCity: "Reims",
    ShipCountry: "France",
    ShipName: "Vins et alcools Chevalier",
    ShipPostalCode: "51100",
    ShipRegion: null,
    ShippedDate: { $date: "1996-07-16T00:00:00.000Z" },
    Shipper: {
      ShipperID: 3,
      CompanyName: "Federal Shipping",
    },
  },
  {
    _id: { $oid: "63a060b9bb3b972d6f4e1fc7" },
    Customer: {
      CustomerID: "TOMSP",
      CompanyName: "Toms Spezialitäten",
      Country: "Germany",
    },
    Employee: {
      EmployeeID: 6,
      LastName: "Suyama",
      FirstName: "Michael",
    },
    Freight: 11.61,
    OrderDate: { $date: "1996-07-05T00:00:00.000Z" },
    OrderID: 10249,
    Orderdetails: [
      {
        UnitPrice: 18.6,
```



```

    Quantity: 9,
    Discount: 0,
    product: {
      ProductID: 14,
      ProductName: "Tofu",
      QuantityPerUnit: "40 - 100 g pkgs.",
    },
    TotalValue: 167.4,
  },
  {
    UnitPrice: 42.4,
    Quantity: 40,
    Discount: 0,
    product: {
      ProductID: 51,
      ProductName: "Manjimup Dried Apples",
      QuantityPerUnit: "50 - 300 g pkgs.",
    },
    TotalValue: 1696,
  },
],
RequiredDate: { $date: "1996-08-16T00:00:00.000Z" },
ShipAddress: "Luisenstr. 48",
ShipCity: "Münster",
ShipCountry: "Germany",
ShipName: "Toms Spezialitäten",
ShipPostalCode: "44087",
ShipRegion: null,
ShippedDate: { $date: "1996-07-10T00:00:00.000Z" },
Shipper: {
  ShipperID: 1,
  CompanyName: "Speedy Express",
},
},
];

```

Zadanie 4

- zmień (popraw) strukturę dokumentu w kolekcji `orders_tmp`
- stwórz obiekt/dokument grupujący inf. o adresie wysyłki
 - np.

```

Shipment : {
  "ShipAddress": "Luisenstr. 48",
  "ShipCity": "Münster",
  "ShipCountry": "Germany",
  "ShipName": "Toms Spezialitäten",
  "ShipPostalCode": "44087",
  "ShipRegion": null,
}

```

- stwórz obiekt/dokument grupujący inf. o datach (zamówienia, wysyłki ...)
 - np.

```

Dates : {
  "OrderDate": { "$date": "1996-07-04T00:00:00.000Z" },
  "RequiredDate": { "$date": "1996-08-01T00:00:00.000Z" },
  "ShippedDate": { "$date": "1996-07-16T00:00:00.000Z" },
}

```

- Dodaj inf. o pełnej wartości całego zamówienia

```
TotalOrderValue : ...
```

Zadanie 4 - rozwiązanie

Wyniki:

przykłady, kod, zrzuty ekranów, komentarz ...

```

use north0;

db.orders_tmp.aggregate([
  {
    $addFields: {
      Shipment: {
        ShipAddress: "$ShipAddress",
        ShipCity: "$ShipCity",

```

```
    ShipCountry: "$ShipCountry",
    ShipName: "$ShipName",
    ShipPostalCode: "$ShipPostalCode",
    ShipRegion: "$ShipRegion"
  },

  Dates: {
    OrderDate: "$OrderDate",
    RequiredDate: "$RequiredDate",
    ShippedDate: "$ShippedDate"
  },

  TotalOrderValue: {
    $round: [
      { $sum: "$Orderdetails.TotalValue" },
      2
    ]
  }
}
},
{
  $project: {
    ShipAddress: 0,
    ShipCity: 0,
    ShipCountry: 0,
    ShipName: 0,
    ShipPostalCode: 0,
    ShipRegion: 0,

    OrderDate: 0,
    RequiredDate: 0,
    ShippedDate: 0
  }
},
{
  $out: "orders_tmp"
}
});
```

Sprawdzenie wyników:

```
db.orders_tmp.find().limit(2);
```

W zadaniu zmieniono strukturę dokumentów w kolekcji `orders_tmp`. Pola związane z adresem wysyłki zostały zgrupowane w obiekcie `Shipment`, a pola związane z datami w obiekcie `Dates`.

Dodatkowo dodano pole `TotalOrderValue`, które przechowuje pełną wartość zamówienia. Wartość ta jest obliczana jako suma pól `TotalValue` ze wszystkich pozycji zamówienia znajdujących się w tablicy `Orderdetails`.